



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1

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Functions in C

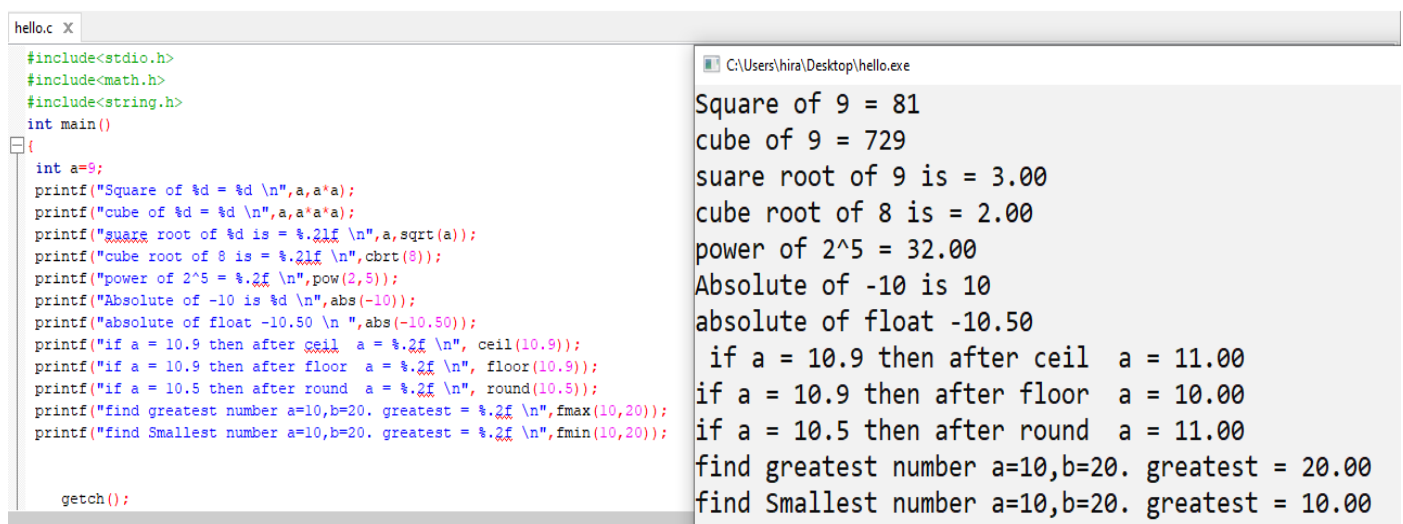
2. <math.h>

Used for mathematical calculations.

Name	Function Use
sqrt()	Square root
Cbrt()	Cube root
pow(x,y)	Power
abs(x)	Absolute value (convert -ve to +ve)
Fabs()	Absolute value for float
Fmax(x,y)	Maximum value return → valid 2 number
Fmax(Fmax(x,y),c)	Maximum value return → valid 3 number
Fmax(fmax(a,b),fmax(c,d))	Maximum value return → valid 4 number
Fmin(x,y)	Minimum value return
ceil(x)	Round up – upper integer
floor(x)	Round down – lower integer
Round()	Round off
sin(x)	Sin (x will be insert in radian value not degree) $\text{Radian} = \text{degree} \times \frac{\pi}{180} \quad (\pi = 3.141)$
cos(x)	Cosine
tan(x)	Tangent
log(x)	Natural logarithm
log10(x)	Log base 10
exp(x)	Exponential
fmod(x,y)	Remainder
hypot(x,y)	hypotenuse

Q. Write a program in c use all function use in <math.h> header file.

```
#include<stdio.h>
#include<math.h>
int main()
{
    int a=9;
    printf("Square of %d = %d \n",a,a*a);
    printf("cube of %d = %d \n",a,a*a*a);
    printf("suare root of %d is = %.2lf \n",a,sqrt(a));
    printf("cube root of 8 is = %.2lf \n",cbrt(8));
    printf("power of 2^5 = %.2f \n",pow(2,5));
    printf("Absolute of -10 is %d \n",abs(-10));
    printf("absolute of float -10.50 \n ",abs(-10.50));
    printf("if a = 10.9 then after ceil  a = %.2f \n", ceil(10.9));
    printf("if a = 10.9 then after floor  a = %.2f \n", floor(10.9));
    printf("if a = 10.5 then after round  a = %.2f \n", round(10.5));
    printf("find greatest number a=10,b=20. greatest = %.2f \n",fmax(10,20));
    printf("find Smallest number a=10,b=20. greatest = %.2f \n",fmin(10,20));
    getch();
    return 0;
}
```



The screenshot displays a C program in a text editor and its execution output. The source code, saved as 'hello.c', includes headers for stdio, math, and string, and uses various mathematical functions. The output window shows the results of these functions for specific inputs.

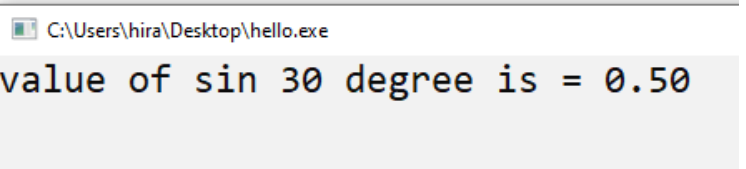
```
hello.c X
#include<stdio.h>
#include<math.h>
#include<string.h>
int main()
{
    int a=9;
    printf("Square of %d = %d \n",a,a*a);
    printf("cube of %d = %d \n",a,a*a*a);
    printf("suare root of %d is = %.2lf \n",a,sqrt(a));
    printf("cube root of 8 is = %.2lf \n",cbrt(8));
    printf("power of 2^5 = %.2f \n",pow(2,5));
    printf("Absolute of -10 is %d \n",abs(-10));
    printf("absolute of float -10.50 \n ",abs(-10.50));
    printf("if a = 10.9 then after ceil  a = %.2f \n", ceil(10.9));
    printf("if a = 10.9 then after floor  a = %.2f \n", floor(10.9));
    printf("if a = 10.5 then after round  a = %.2f \n", round(10.5));
    printf("find greatest number a=10,b=20. greatest = %.2f \n",fmax(10,20));
    printf("find Smallest number a=10,b=20. greatest = %.2f \n",fmin(10,20));
    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
Square of 9 = 81
cube of 9 = 729
suare root of 9 is = 3.00
cube root of 8 is = 2.00
power of 2^5 = 32.00
Absolute of -10 is 10
absolute of float -10.50
if a = 10.9 then after ceil  a = 11.00
if a = 10.9 then after floor  a = 10.00
if a = 10.5 then after round  a = 11.00
find greatest number a=10,b=20. greatest = 20.00
find Smallest number a=10,b=20. greatest = 10.00
```

❖ Find trigonometry value

```
#include<conio.h>
#include<math.h>
int main()
{
    printf("value of sin 30 degree is = ");
    int d=30;
    float r,f;
    r=d*3.14/180; // convert degree to radian
    f=sin(r);
    printf("%.2f", f);
    getch();
}
```



3. string.h

Used to work with text data (sequence of characters). It's much easier than C-style character arrays.

Name	Function use								
Strlen()	Length of string								
Strcpy()	Copy string Syntax → strcpy(b,a); // strcpy(target , original) OR strcpy(destination, source)								
Strncpy()	Copy limited characters Syntax → strncpy(destination, source, n);								
Strcmp()	Compare string. Syntax - strcmp(string1, string2);								
	Rule								
	<table><tr><th>Condition</th><th>Result</th></tr><tr><td>Str1 == str 2</td><td>0</td></tr><tr><td>Str1 < str2</td><td>-ve number</td></tr><tr><td>Str1 > str2</td><td>+ve number</td></tr></table>	Condition	Result	Str1 == str 2	0	Str1 < str2	-ve number	Str1 > str2	+ve number
	Condition	Result							
	Str1 == str 2	0							
Str1 < str2	-ve number								
Str1 > str2	+ve number								
Strcat()	Concatenate or Add two strings Syntax - strcat(destination, source);								
Strcat()	Concatenate limited characters								
Strchr()	Find character→ print after all string when find character								

	Syntax - strchr(string, character);
Strstr()	Find substring

Q. Write a program use all function available in <string.h>.

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
int main()
{
    char c[20] = "biharsharif";
    char x[20],s[20];
    printf("String Length of c[20]Biharsharif is = %d \n",strlen(c));
    // OR
    printf("String length of Hira is = %d \n", strlen("hira"));
    strcpy(x,c); //strcpy(target , original)
    printf("value of x[20] is = %s \n",x);
    strncpy(s,c,5); //strncpy(destination, source, n);
    printf("Limited string copy s[] = %s \n",s);
    //OR
    strncpy(s,"Hirakumar",4);
    s[4]='\0'; // remove null character
    printf("limited String print = %s \n",s);
    printf("Find character in string = %s \n",strchr(c,'r'));
    strcat(s,c);
    printf("add two string is = %s \n",s);
    printf("what c & s is = %d",strcmp(c,s)); // compair string

    getch();
}
```

```

}
hello.c X
int main()
{
    char c[20] = "biharsharif";
    char x[20], s[20];
    printf("String Length of c[20]Biharsharif is = %d \n", strlen(c));
    // OR
    printf("String length of Hira is = %d \n", strlen("hira"));
    strcpy(x, c); //strcpy(target, original)
    printf("value of x[20] is = %s \n", x);
    strncpy(s, c, 5); //strncpy(destination, source, n);
    printf("Limited string copy s[] = %s \n", s);
    //OR
    strncpy(s, "Hirakumar", 4);
    s[4] = '\0'; // remove null character
    printf("limited String print = %s \n", s);
    printf("Find character in string = %s \n", strchr(c, 'r'));
    strcat(s, c);
    printf("add two string is = %s \n", s);
    printf("what c & s is = %d", strcmp(c, s)); // compare string

    getch();
}

```

C:\Users\hira\Desktop\hello.exe

String Length of c[20]Biharsharif is = 11
String length of Hira is = 4
value of x[20] is = biharsharif
Limited string copy s[] = bihar
limited String print = Hira
Find character in string = rsharif
add two string is = Hirabiharsharif
what c & s is = 1

4. <ctype.h>

Used for character testing and conversion.

name	Function use
isalpha()	Alphabet check
isdigit()	Digit check
isalnum()	Alphanumeric check
islower()	Lower case check
isupper()	Upper case check
isspace()	Space check
toupper()	Convert to uppercase
tolower()	Convert to lowercase

5. <fstream>

Used for file handling.

Class	Use
ofstream	In file write (data insert)
ifstream	File read (get/collect data)
fstream	Read & Write both

name	Function use	syntax
open()	Open file	
close()	Close file	
read()	Read from file	
write()	Write to file	
getline()	Read full line	
eof()	End of file check	

seekg()	Move get pointer	
seekp()	Move put pointer	
tellg()	Get position	
tellp()	Get position	

6. <algorithm>

Used for sorting and searching algorithms.

7. <string>

Used to work with string class.

8. <vector>

Used for dynamic arrays.

9. <iomanip>

Used for formatted output. E.g. – alignment, fixed decimal,

10. <cstdlib>

Used for utility functions. E.g – exit(), free(), rand(), malloc(), ...

And so many,